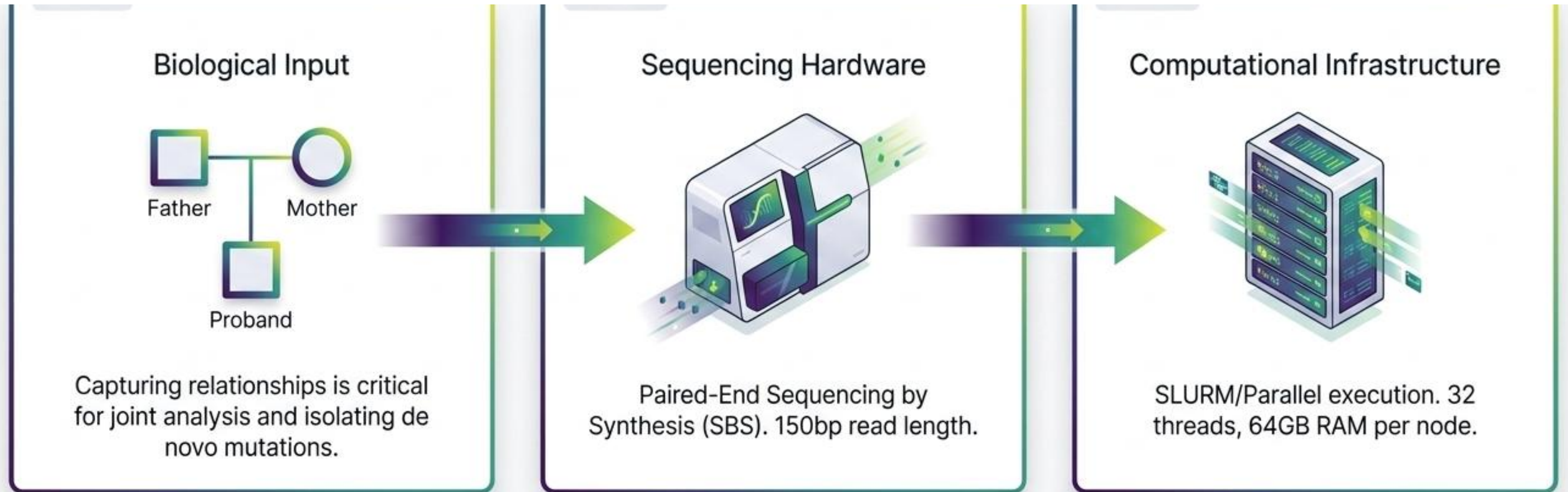


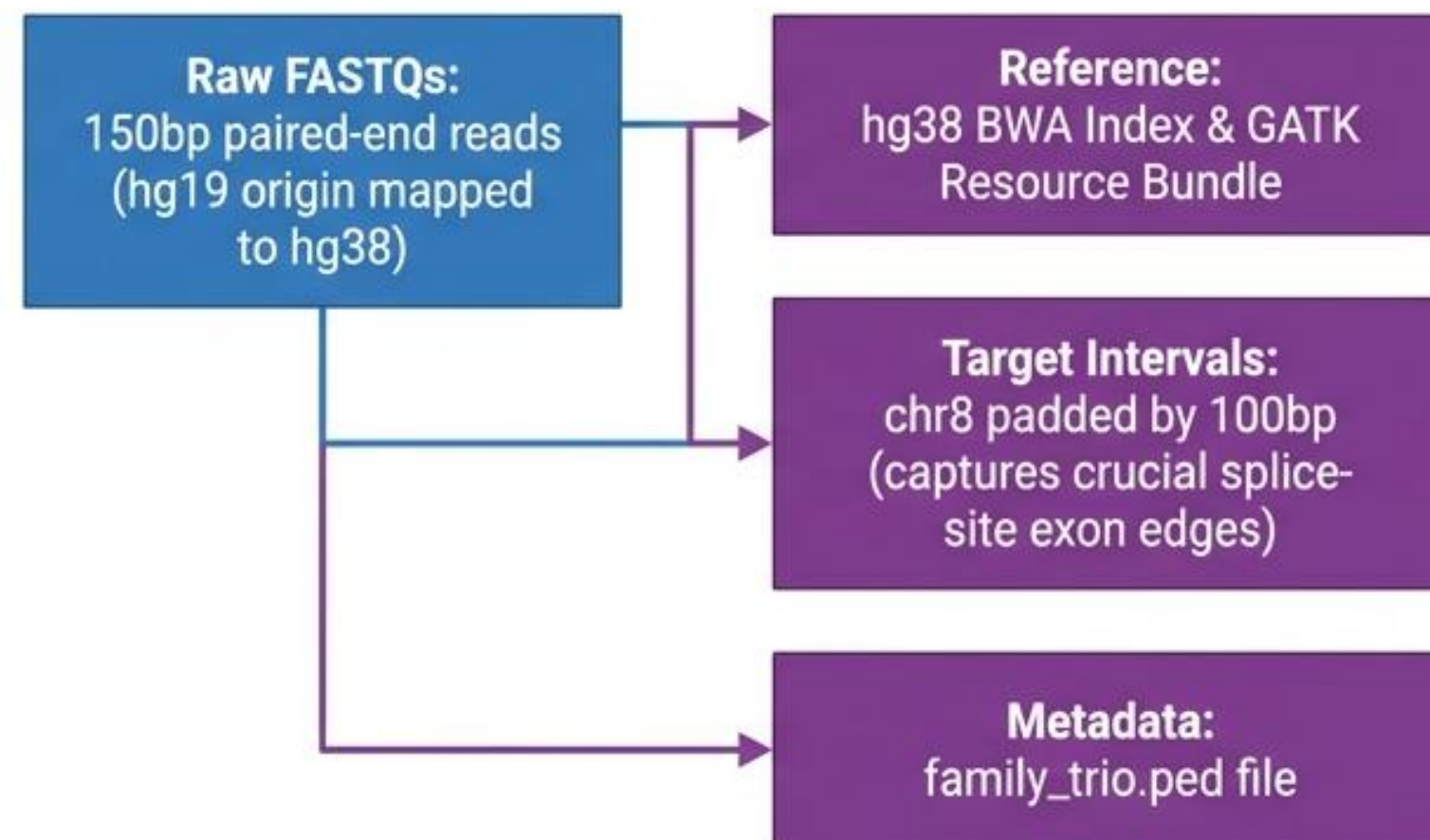
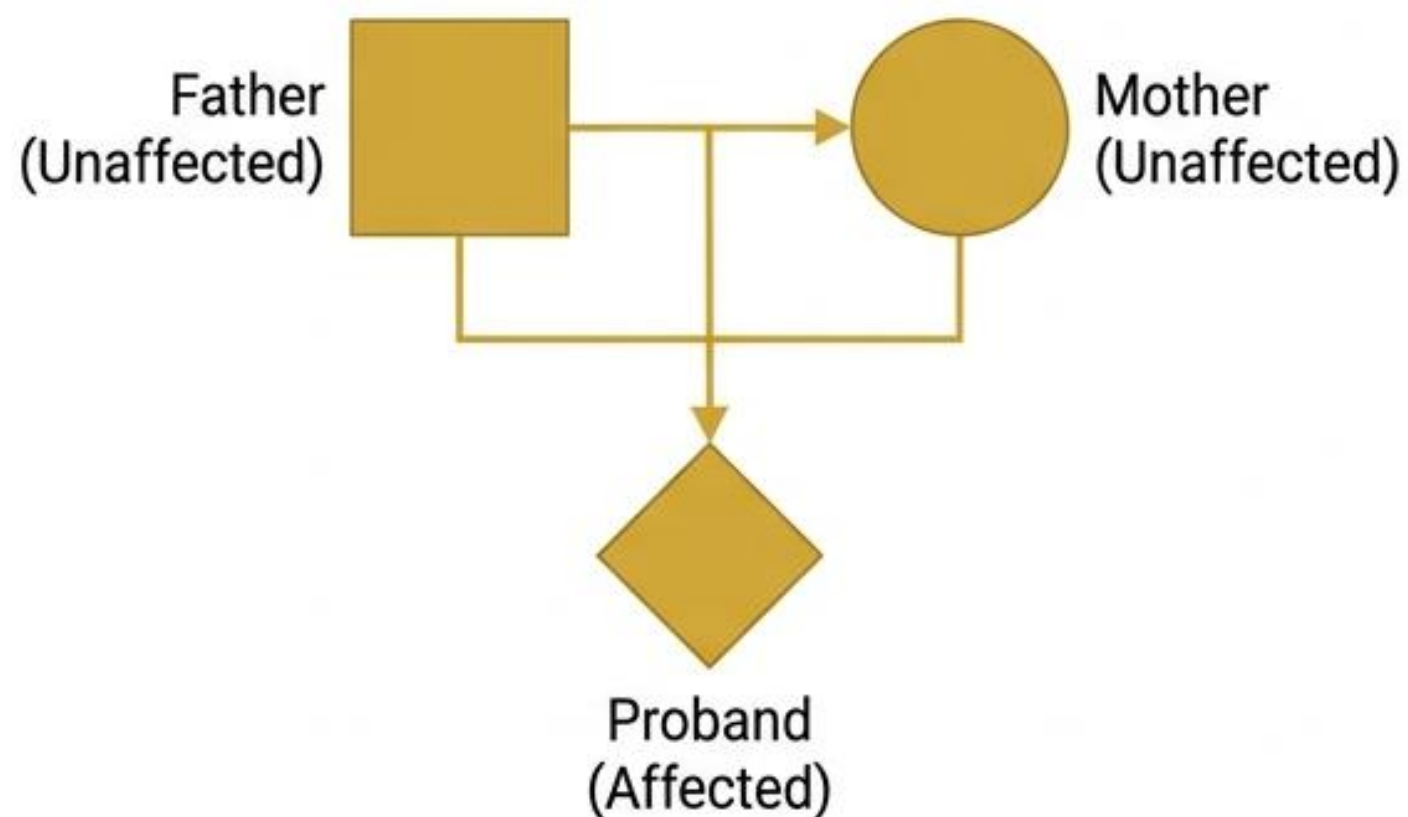
# Genome Sequencing and Analysis



```
REFERENCE_GENOME = "GRCh38/hg38"
```

Consistency in the reference sequence dictionary is **absolutely mandatory** across the entire pipeline.

[ EXPERIMENTAL DESIGN ]



We sequence trios rather than isolated individuals to leverage genetic inheritance rules. A pedigree file (.ped) allows downstream joint-genotyping to validate weak variant signals in a proband if present in a parent, and confidently identify de novo mutations responsible for rare diseases.



# The Variant Discovery Matrix

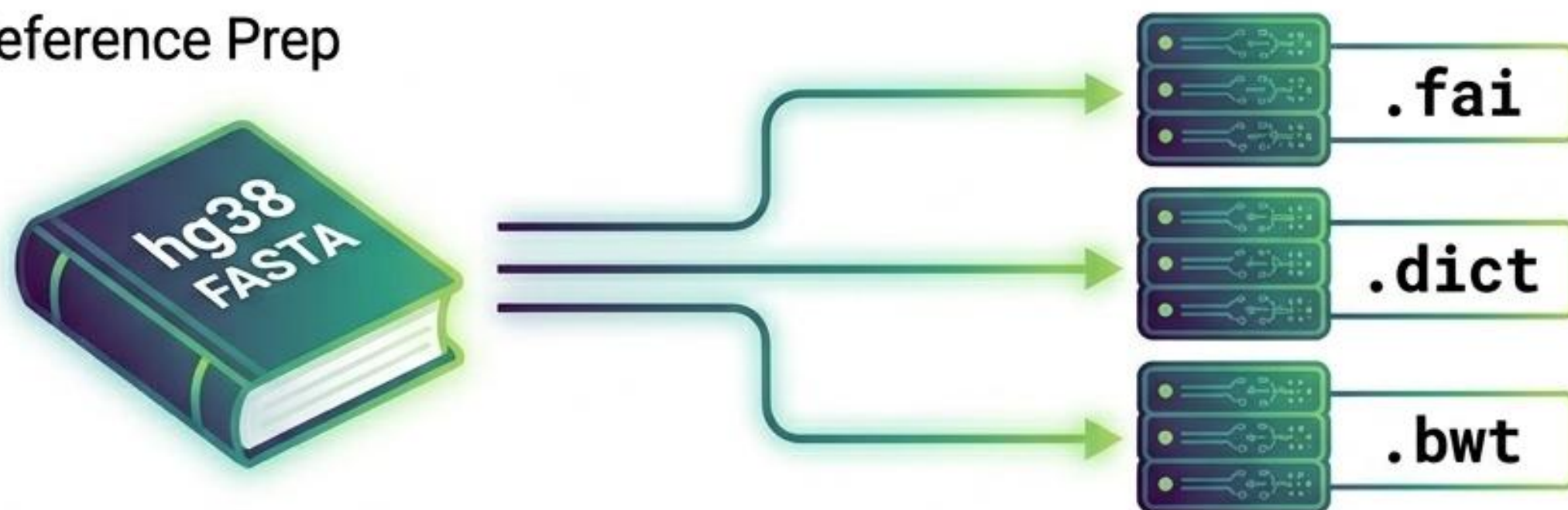
|                       | Germline Discovery                                          | Somatic Discovery                                                                         |
|-----------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Biological Origin     | Inherited variants present in all cells.                    | Acquired mutations, e.g., in tumours.                                                     |
| State & Pliancy       | Focuses on stable diploid states (heterozygous/homozygous). | Focuses on clonal/subclonal mixtures.                                                     |
| Depth Requirements    | ~30x minimum coverage.                                      | >90x ultra-deep coverage required.                                                        |
| Pipeline Optimisation | Tuned for high specificity.                                 | Tuned for high sensitivity to detect low-allele-frequency variants against normal tissue. |

## Trimming Adapters



Adapters create false positive SNPs at read edges. fastp applies a sliding 4bp window, trimming when quality drops below Q20.

## Reference Prep



Pre-computing data structures allows rapid querying. Mismatched contig names (e.g., 'chr1' vs '1') will fatally crash the pipeline.



## [ QC & TRIMMING ]

```
fastp \  
  --in1 $R1 --in2 $R2 \  
  --out1 $T1 --out2 $T2 \  
  --detect_adapter_for_pe --cut_right \  
  --length_required 50 \  
  --correction --overrepresentation_analysis \  
  --json $JSON --html $HTML
```

**Purpose:** Remove technical artifacts (adapters) and low-quality 3' base calls that mimic true SNPs.

Input: Raw FASTQs → Outputs: Trimmed FASTQs → QC Reports

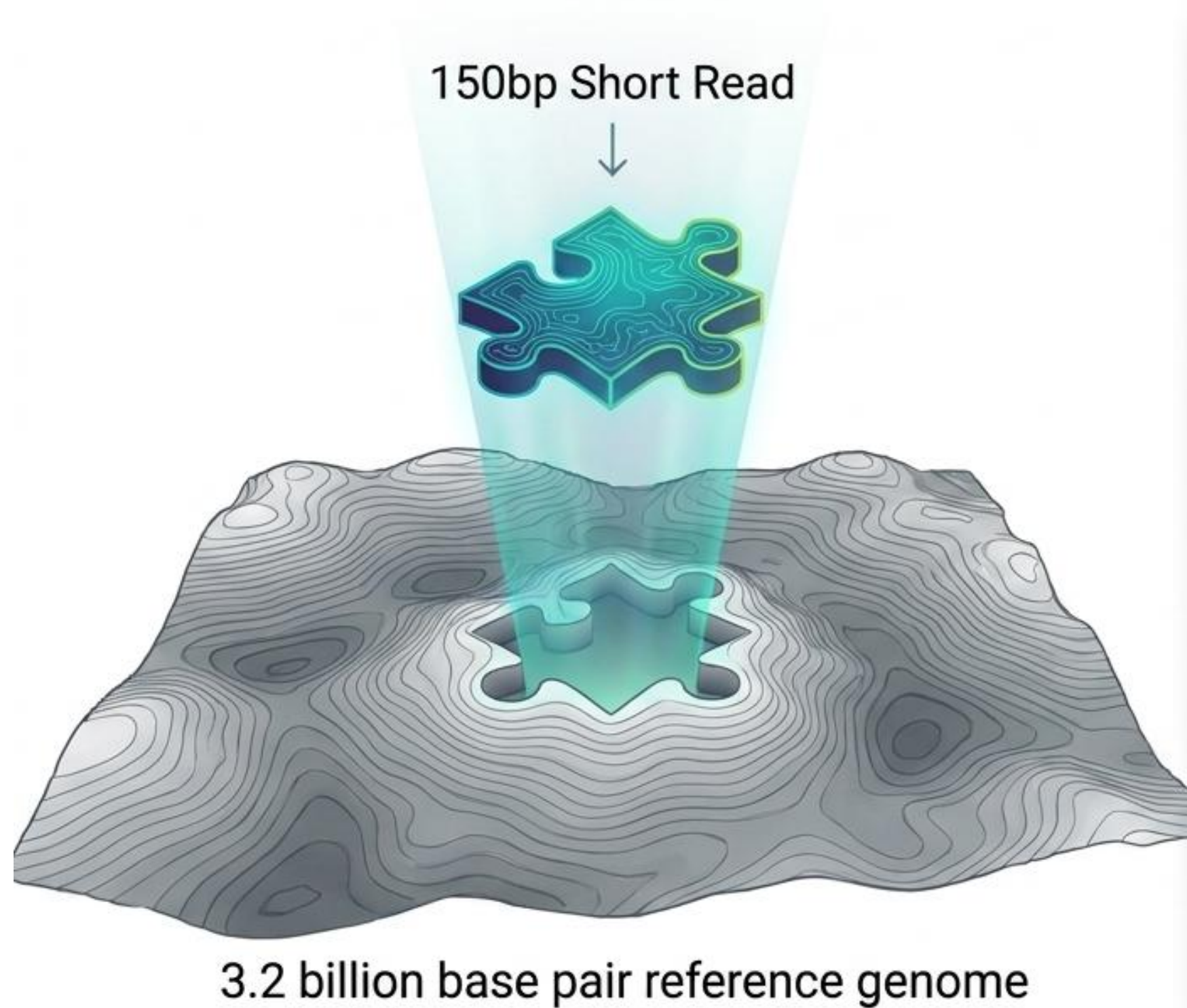
--length\_required 50 is enforced strictly for exome sequencing. Fragments shorter than 50bp map ambiguously across the genome, increasing false positive variant calls.

### FASTQ Record Anatomy

|                           |                        |
|---------------------------|------------------------|
| @SRR123456.1 1 length=150 | ← @Header              |
| GATCTGACCTG               | ← Raw Sequence         |
| +                         | ← Separator            |
| IIIIIIIIIIII              | ← ASCII Quality String |

↑ Phred Score 40 = 99.99% Base Accuracy

# The Alignment Challenge





## [ ALIGNMENT ]

```
RG="@RG\tID:${SAMPLE}.1\tSM:${SAMPLE}\tLB:  
${LIBRARY}_${SAMPLE}\tPL:ILLUMINA"
```

```
bwa mem -t $THREADS -R "$RG" -M $GENOME $R1 $R2 \  
| samtools sort -@ $THREADS -m 4G -O BAM -o  
$OUT_BAM
```

**Purpose:** Locate the precise genomic origin of millions of short reads against the 3.2 billion bp hg38 reference genome.

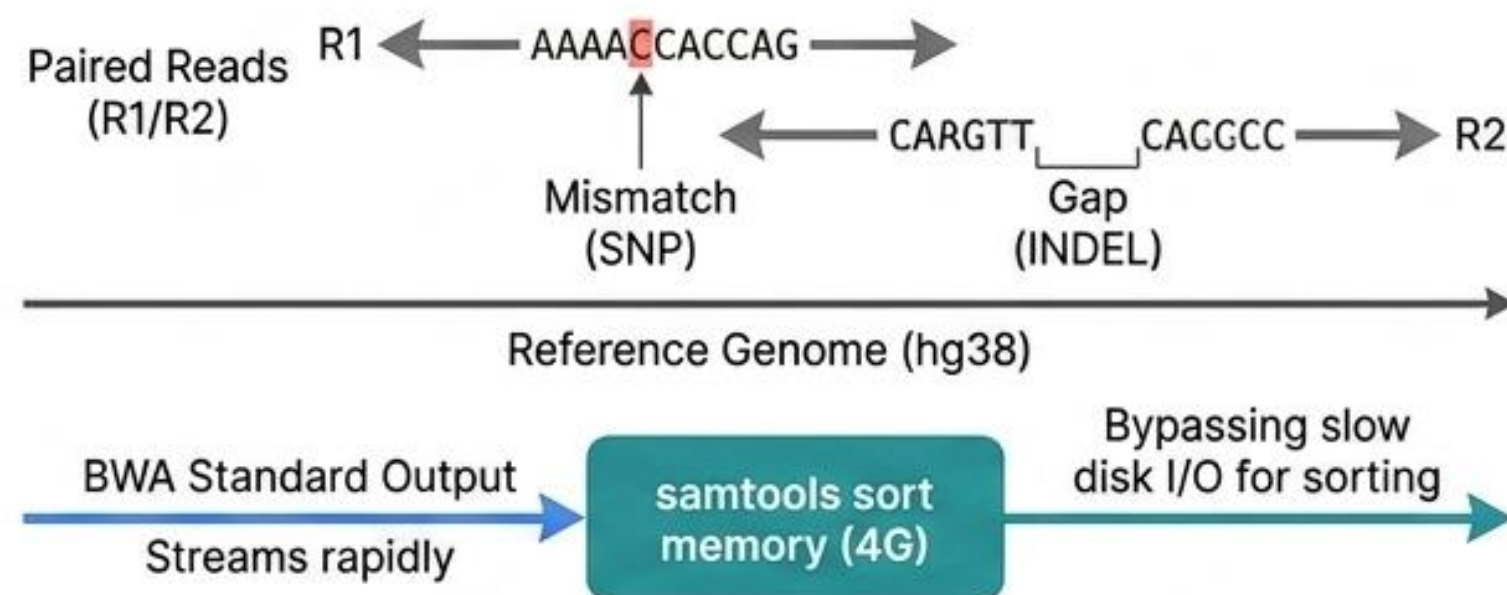
**Inputs:** Trimmed FASTQs

hg38 Index

**Output:** Sorted BAM

-R "\$RG" is **strictly mandatory**. GATK relies entirely on Read Groups (SM=Sample, LB=Library) to distinguish technical batch effects from true biology during variant calling. -M ensures split reads are flagged for Picard.

### Short-Read Mapping Schematic



## [ BAM PROCESSING ]

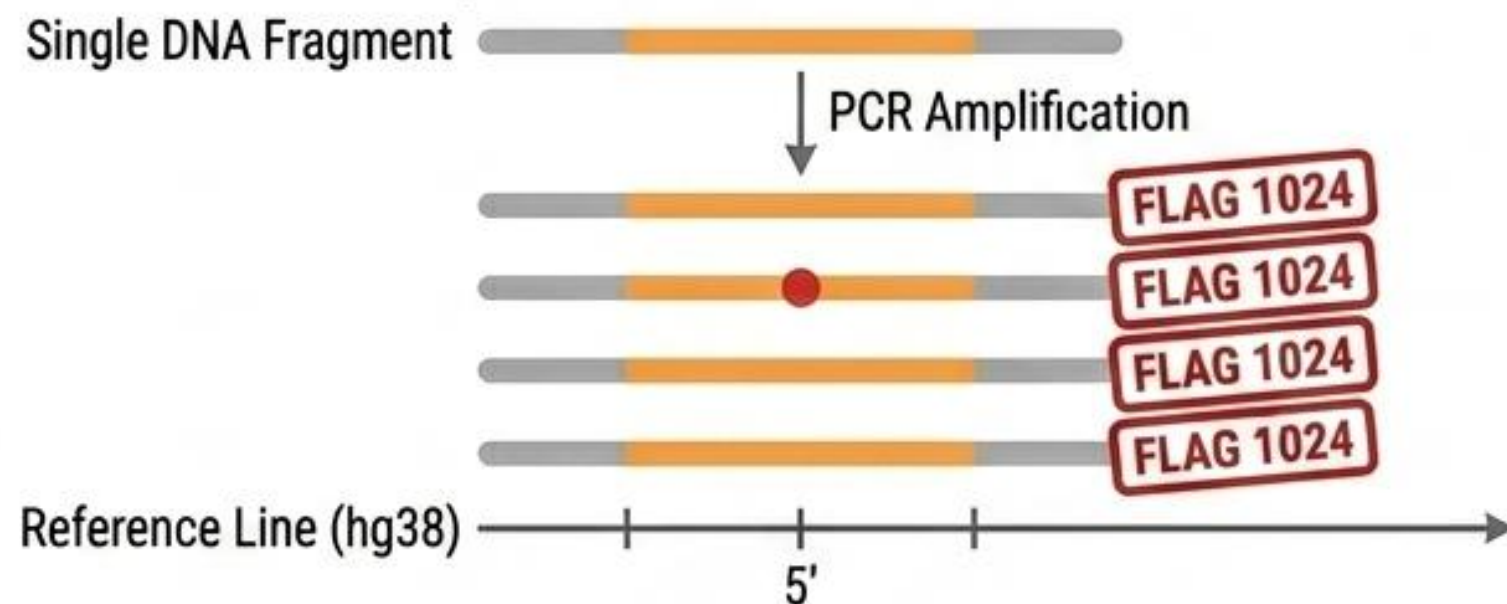
```
gatk MarkDuplicatesSpark \  
  -I $IN_BAM \  
  -O $OUT_BAM \  
  -M $METRICS \  
  --spark-master local[$THREADS] \  
  --java-options "-Xmx${MEM}"
```

**Purpose:** Library preparation requires PCR. A polymerase error on early cycles propagates into thousands of artificial reads. If unflagged, a single machine error masquerades as high-confidence homozygous biology.

Input: Sorted BAM → Output: MarkDup BAM + Metrics

**MarkDuplicatesSpark** utilizes Apache Spark in local mode for multi-threaded horizontal scaling, drastically reducing WGS runtime.

### Duplicate Read Visualization

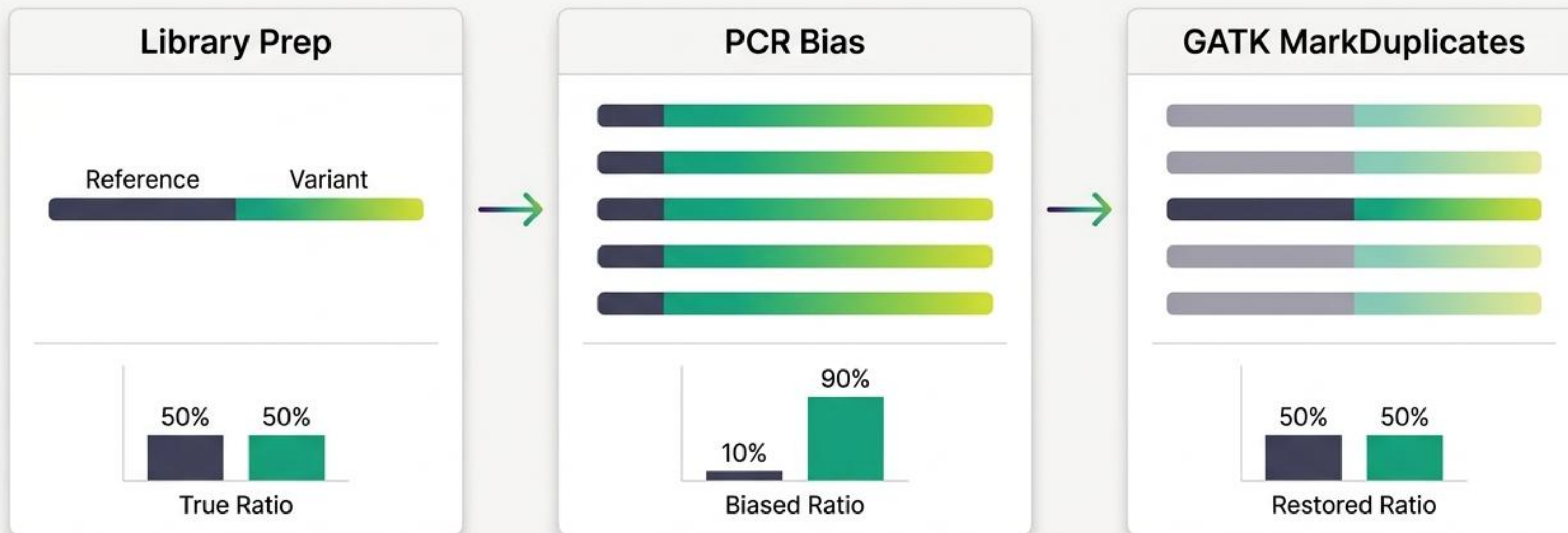


Duplicates are marked to preserve data integrity, not physically deleted.



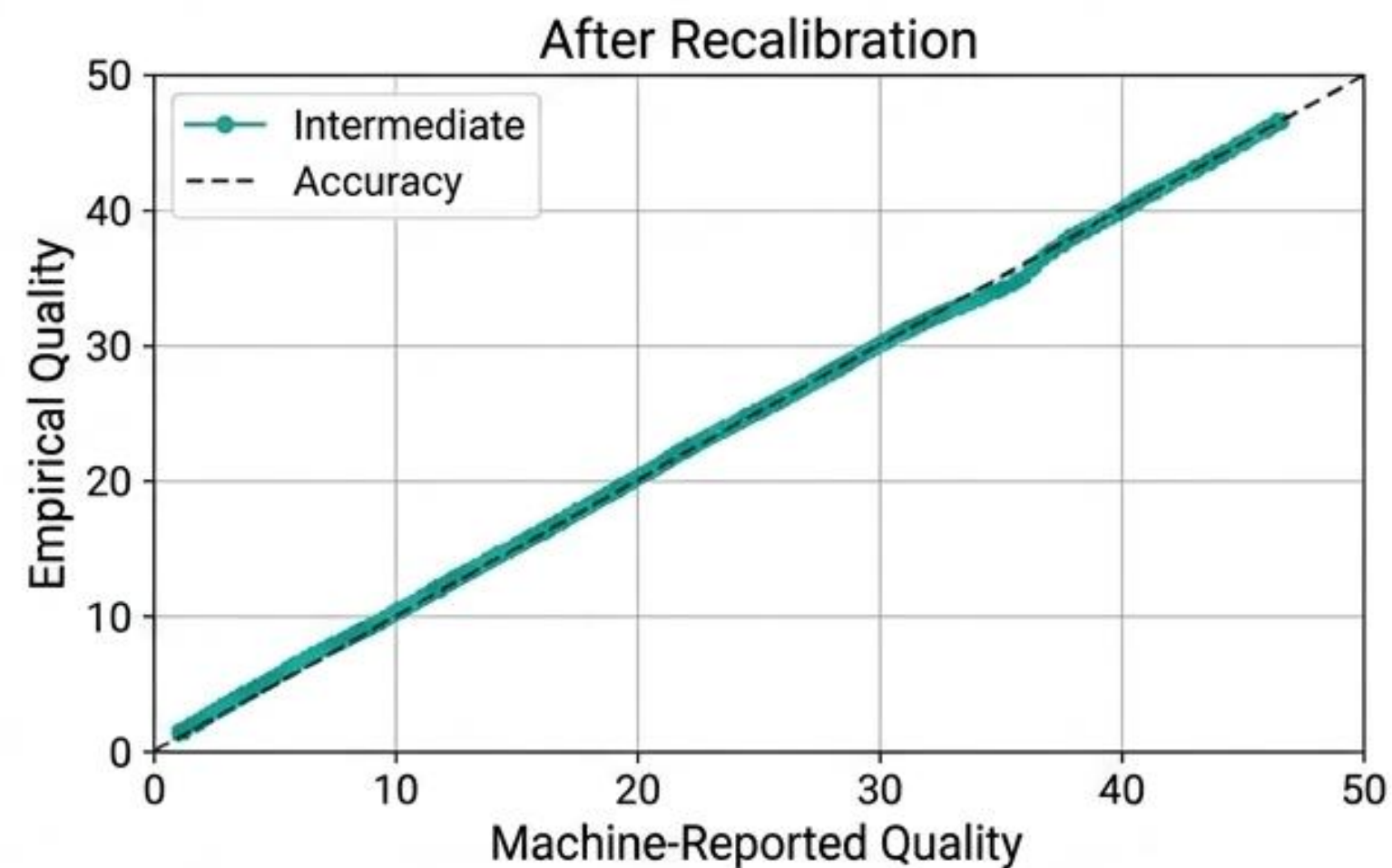
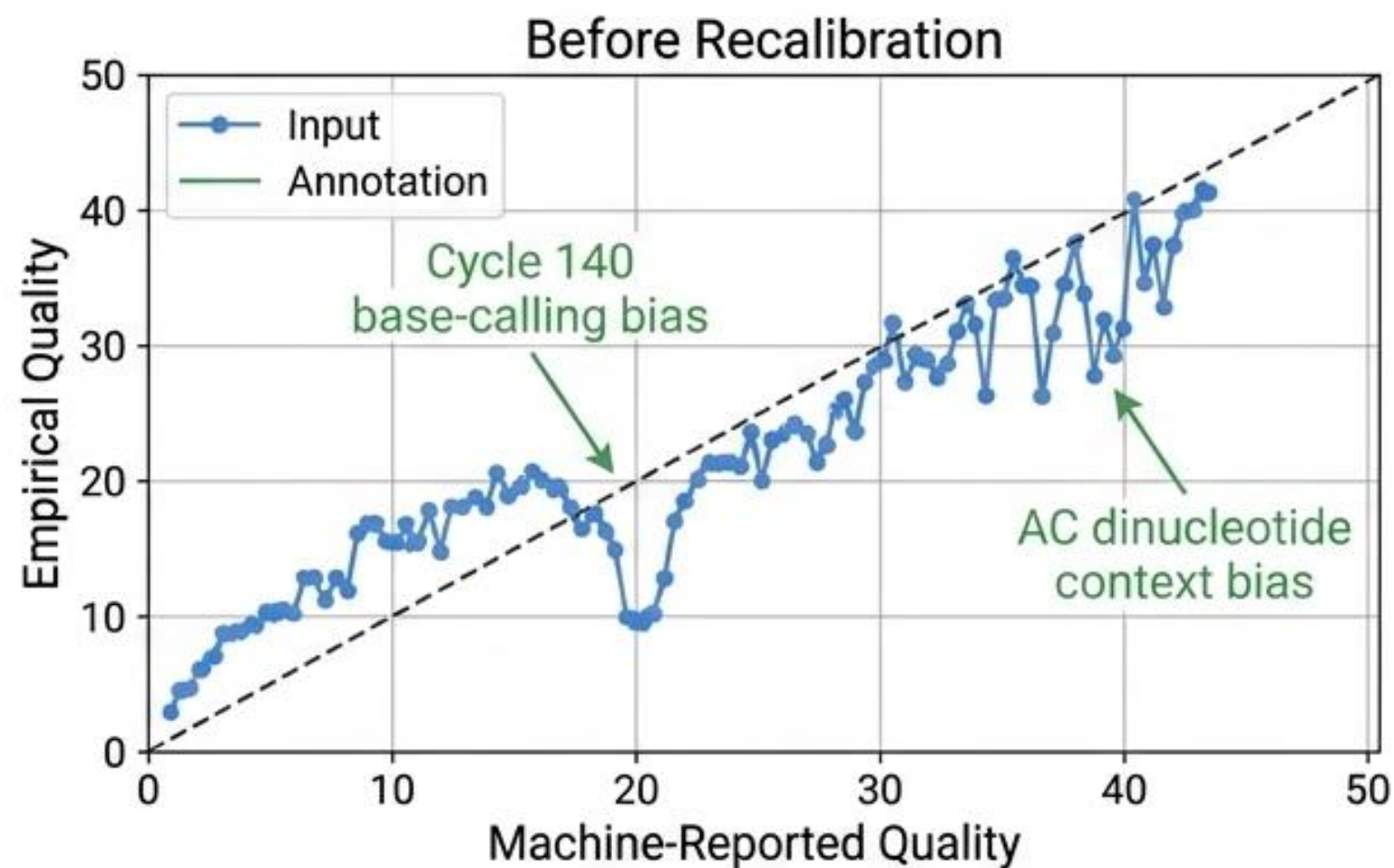
# Mitigating PCR Amplification Bias

Why MarkDuplicates is essential for statistical confidence.



GATK prefers duplicates marked, not deleted. This allows the engine to track total coverage dynamics while ignoring them only during active variant calling.

## [ BQSR CONCEPT ]



Sequencers systematically over- and under-estimate base qualities based on their physical position on the flowcell, the read cycle, and surrounding sequence context. Base Quality Score Recalibration (BQSR) uses machine learning to build an empirical error model and correct these scores, ensuring downstream variant callers make statistical decisions based on truth, not physics.



```
gatk BaseRecalibrator \  
  -I $IN_BAM -R $GENOME \  
  --known-sites $DBSNP \  
  --known-sites $MILLS \  
  --known-sites $KG_INDELS \  
  -O $RECAL_TABLE  
  
gatk ApplyBQSR \  
  -I $IN_BAM \  
  --bqsr-recal-file $RECAL_TABLE \  
  -O $OUT_BAM
```

**Purpose:** Phase 1 analyzes covariation to build the empirical error model; Phase 2 rewrites the BAM file's base quality scores.



### The Biological Rationale for --known-sites

The algorithm mathematically assumes ALL mismatches against the reference genome are sequencing errors. Therefore, we must pass dbSNP and 1000 Genomes databases to mask known human genetic variation.



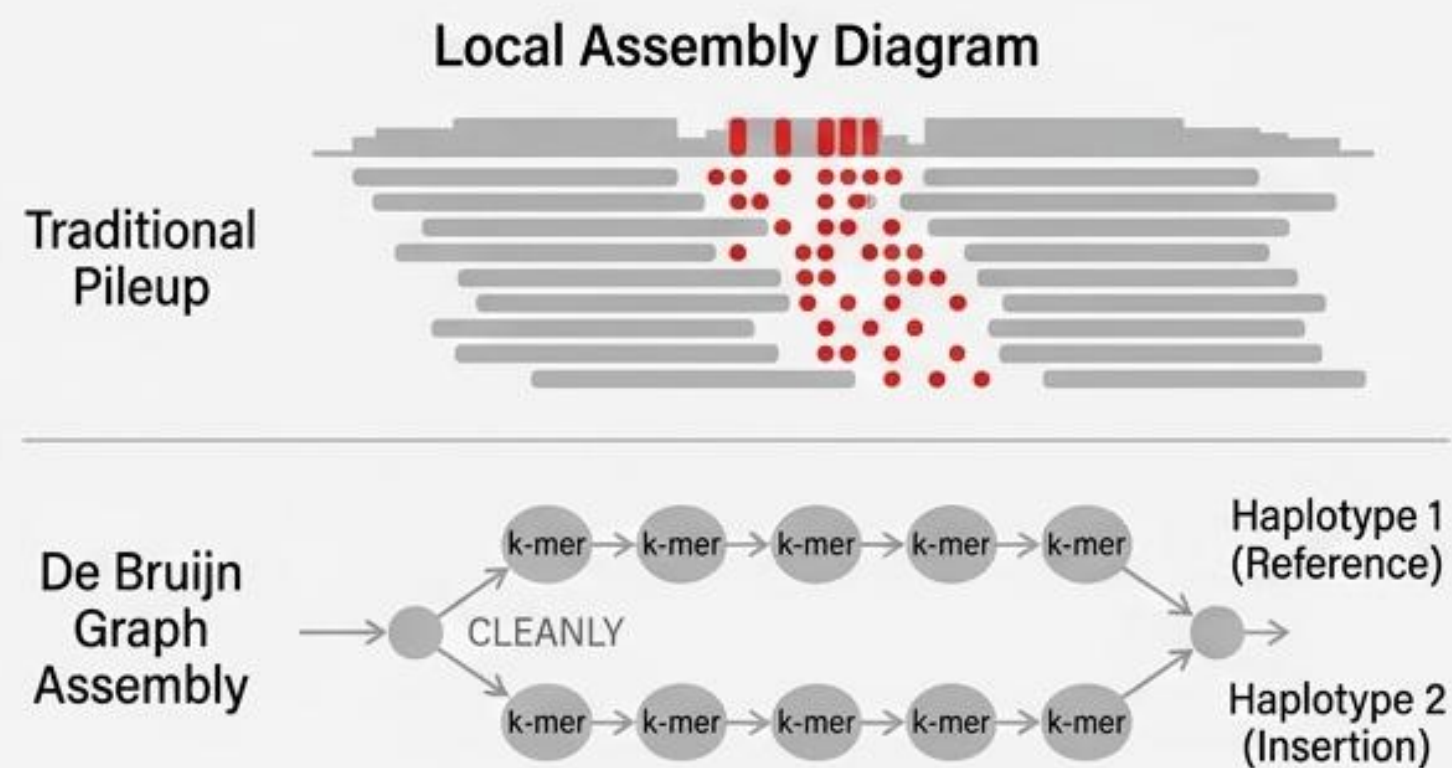
```
gatk HaplotypeCaller \
  -R $GENOME -I $IN_BAM \
  -L $INTERVALS -ERC GVCF \
  --pcr-indel-model NONE \
  --dbsnp $DBSNP \
  -O $OUT_GVCF
```

**Purpose:** Identify SNPs and INDELs. Instead of just counting vertically mapped reads (which fails catastrophically around INDELs), it discards the alignment in active regions and performs localized de novo assembly.

Input: BQSR BAM

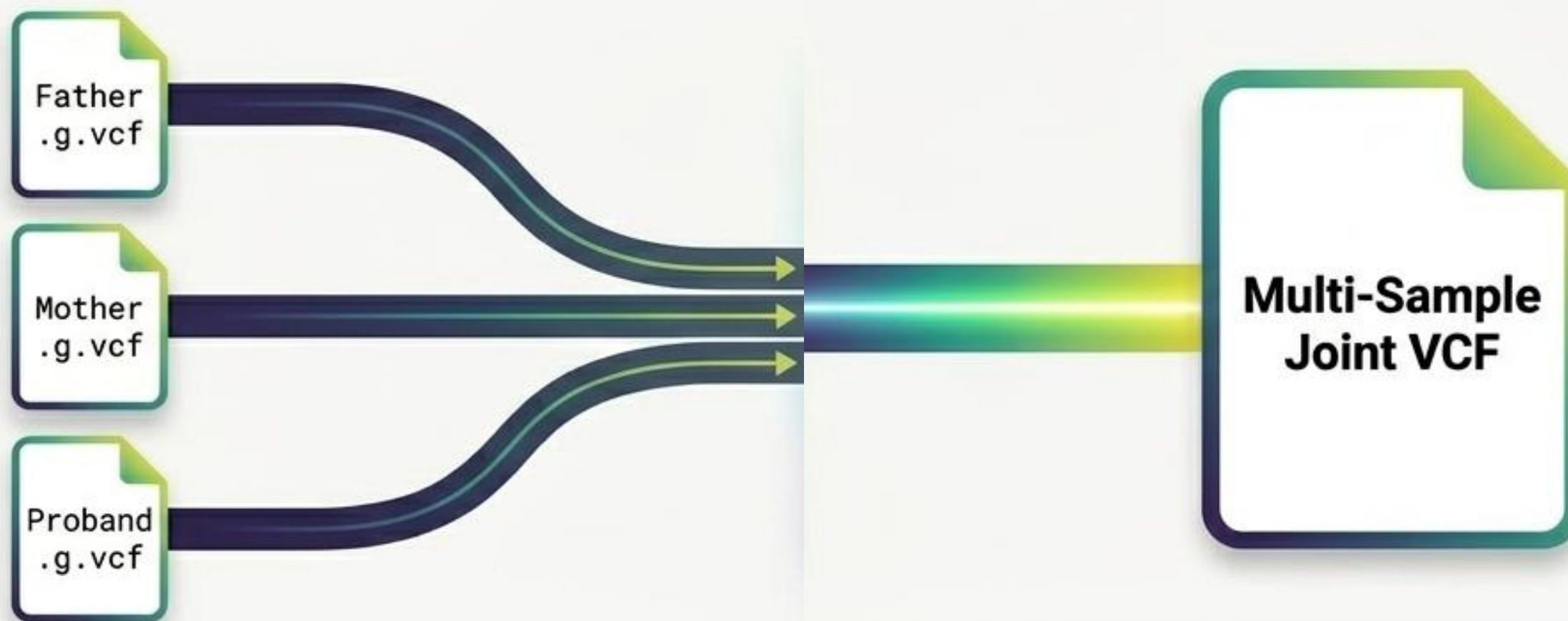
Output: Genomic VCF (GVCF)

Parameter note: -ERC GVCF emits reference confidence blocks for non-variant sites. To merge cohorts later, we must prove a sample actively matched the reference at a position, rather than simply having zero coverage.





# Joint Genotyping



## Biological Rationale

Empowering cohort-wide discovery. A variant with a weak signal in the Proband but a strong signal in the Father can be confidently called in both. The pedigree map is crucial for isolating true de novo variants.

## Computational Note

For cohorts >50 samples, CombineGVCFs scales poorly. Distributed arrays via GenomicsDBImport are required for efficient merging.

## [ JOINT GENOTYPING ]

```
gatk CombineGVCFs \  
  -V father.g.vcf \  
  -V mother.g.vcf \  
  -V proband.g.vcf \  
  -O trio.g.vcf\n
```

```
gatk GenotypeGVCFs \  
  -V trio.g.vcf -R $GENOME \  
  --pedigree $PEDIGREE \  
  -O $JOINT_VCF
```

**Purpose:** Converts probabilities from multiple individual GVCFs into finalized genotype calls (0/1, 1/1) across all samples simultaneously. Leverages population/family data to separate signal from noise.

**Inputs:** Individual GVCFs



**Output:** Joint VCF

**Note:** CombineGVCFs is suitable for trios. For cohorts >50, modern architecture mandates GenomicsDBImport.

### Joint Genotyping Schematic

|         |                                                                                                                |                       |                       |
|---------|----------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|
| Father  | <br>40 reads (ALT allele) | Statistical<br>Rescue | 0/1                   |
| Mother  | <br>0 reads               |                       | 0/0                   |
| Proband | <br>4 reads (ALT allele)  |                       | 0/1<br>(Heterozygous) |



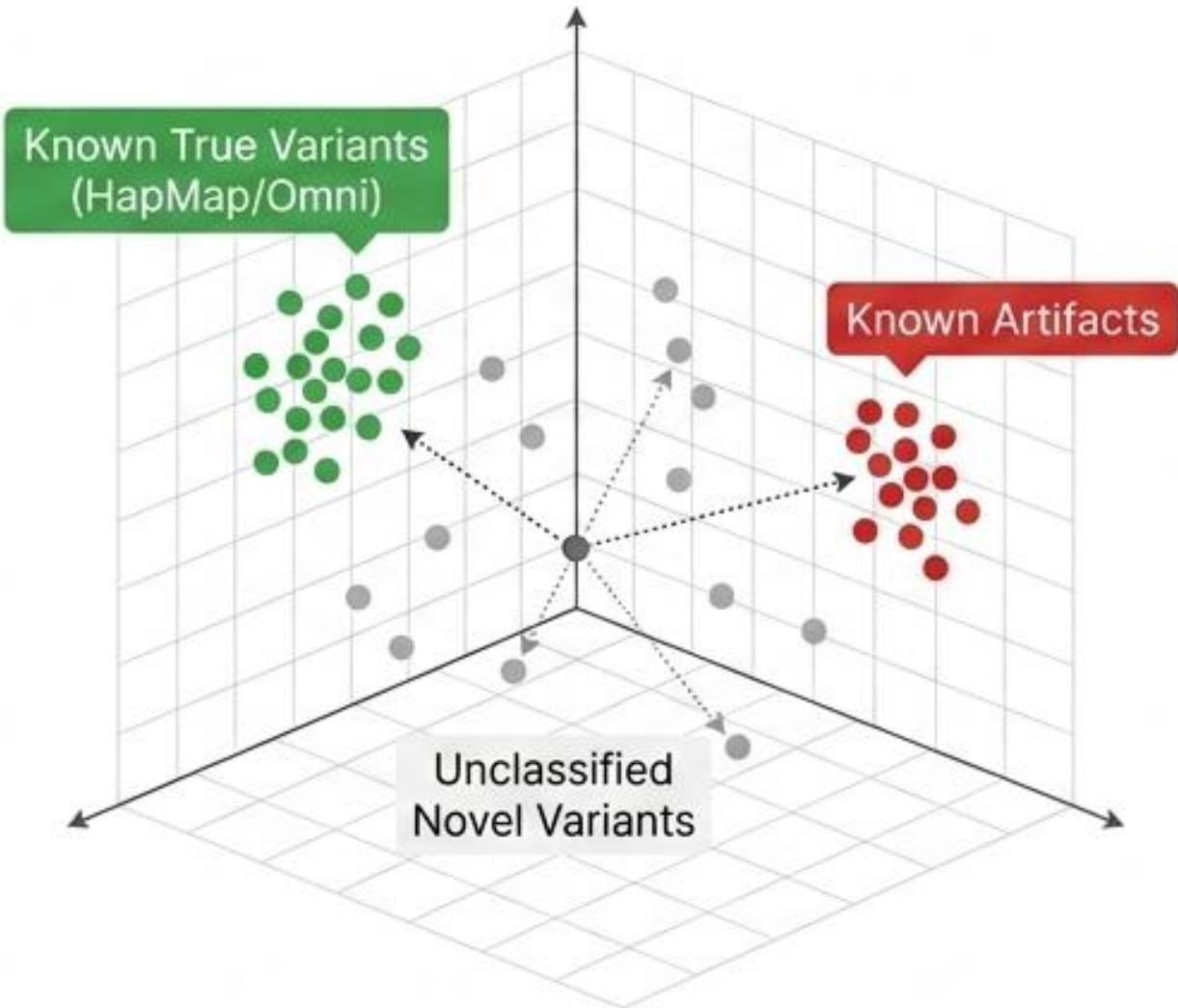
# The Filtration Strategy Matrix

|                   | VQSR<br>(Variant Quality Score Recalibration)                                                                                    | Hard Filtering                                                         |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Mechanism         | Uses a Gaussian mixture model learned from highly validated known SNPs (HapMap/Omni).                                            | Uses fixed unidimensional thresholds (e.g., $QD < 2.0$ , $FS > 60.0$ ) |
| Data Requirements | Requires large cohorts or Whole Genomes to build a robust statistical model.                                                     | Used as a fallback for small datasets or single Exomes.                |
| Key Metric        | <b>Tranches</b> (e.g., A <b>99.5</b> tranche captures 99.5% of known truth variants, discarding the rest to maximise precision). | Absolute cutoffs.                                                      |

## VQSR vs. Hard Filtering

| Dimension        | VQSR<br>(GATK Best Practice)                      | Hard Filtering<br>(Fallback)                 |
|------------------|---------------------------------------------------|----------------------------------------------|
| Methodology      | Gaussian mixture models<br>(Machine Learning)     | Static threshold cutoffs<br>(e.g., QD < 2.0) |
| Dimensionality   | Evaluates 6+ metrics<br>simultaneously            | Operates on rigid<br>1D cutoffs              |
| Data Requirement | Massive data<br>(>30 exomes for INDELs)           | Works on isolated<br>single samples (n=1)    |
| Outcome          | Yields a continuous<br>probability <b>tranche</b> | Results in strict<br>binary <b>PASS/FAIL</b> |

## VQSR Concept Illustration

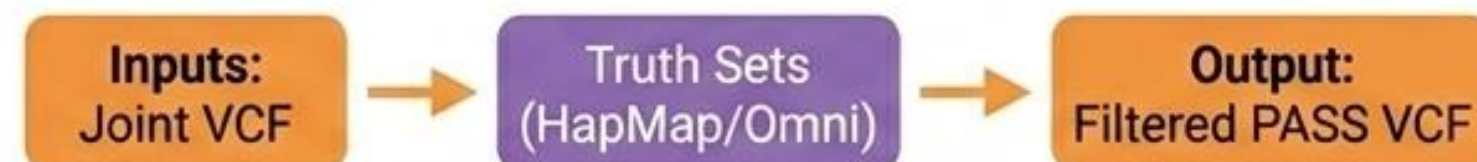




## [ FILTERING EXECUTION ]

```
gatk VariantRecalibrator \  
  -mode SNP \  
  --resource:hapmap,truth=true,training=true $HAPMAP \  
  --resource:omni,truth=false,training=true $OMNI \  
  --resource:dbsnp,known=true $DBSNP \  
  -an QD -an MQ -an FS -an SOR \  
  -O $SNP_RECAL  
  
gatk ApplyVQSR \  
  -mode SNP \  
  --truth-sensitivity-filter-level 99.5 \  
  -O $OUT_VCF
```

**Purpose:** Train a machine learning model on highly validated truth sets to dynamically score and filter out false positives.



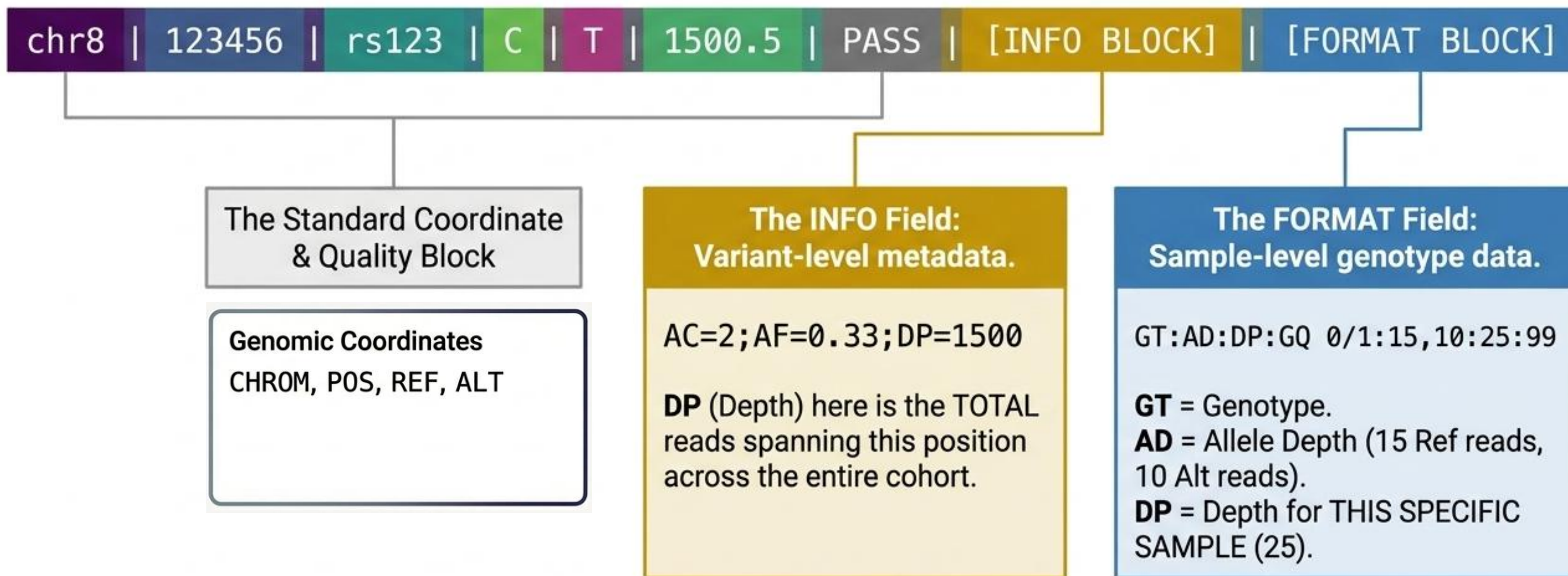
### Parameter Rationale: --truth-sensitivity-filter-level 99.5

The threshold is set dynamically to capture exactly 99.5% of the known true variants present in the training set.

We intentionally sacrifice the bottom 0.5% to ensure a pristine, highly confident final dataset.

INDELs typically use a stricter 99.0 due to intrinsically higher alignment error rates.

## [ VCF STRUCTURE ]





## [ ANNOTATION ]

```
vep --input_file $IN_VCF \  
    --everything --canonical \  
    --af_gnomadg --max_af \  
    --pick_allele \  
    --offline --dir_cache $VEP_CACHE \  
    --fasta $GENOME -o $VEP_VCF
```

**Purpose:** Map isolated coordinate changes (chr8:C>T) to transcript models to predict protein alterations and retrieve critical population frequencies.

**Input:** Filtered VCF

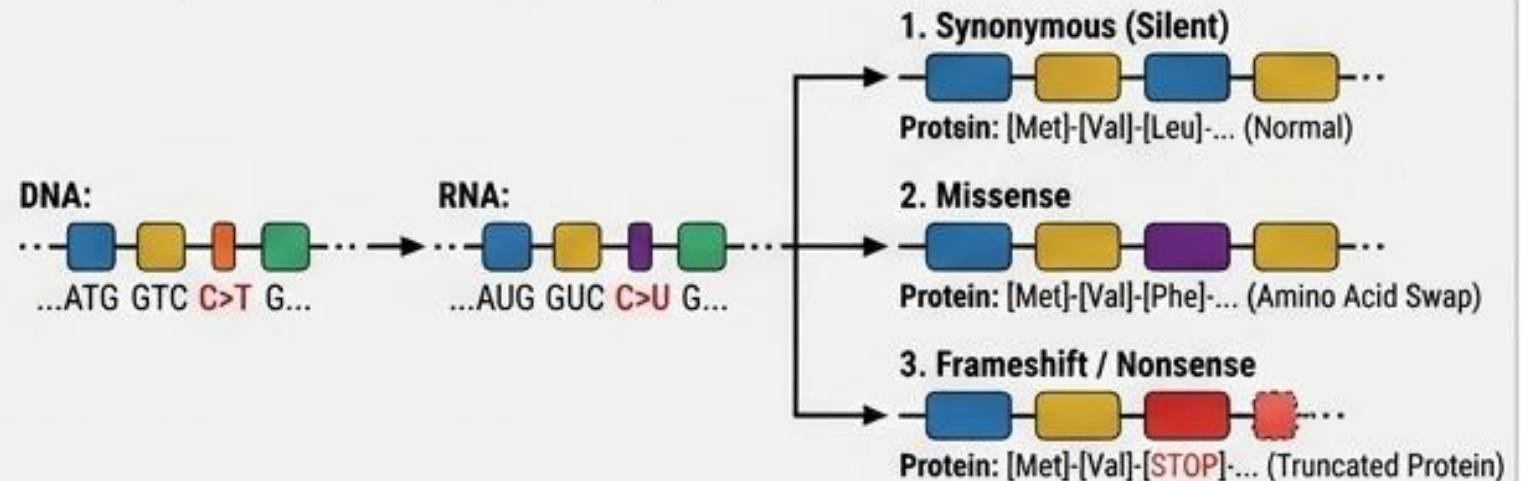
**Ref:** VEP Cache

**Output:** VEP-Annotated VCF

### Parameter highlights:

- `--af_gnomadg` queries the gnomAD database. A variant with a 5% population frequency is almost certainly benign for a rare Mendelian disease.
- `--pick_allele` forces VEP to isolate the single most severe biological consequence to prevent combinatorial explosion.

### Consequence Prediction Map



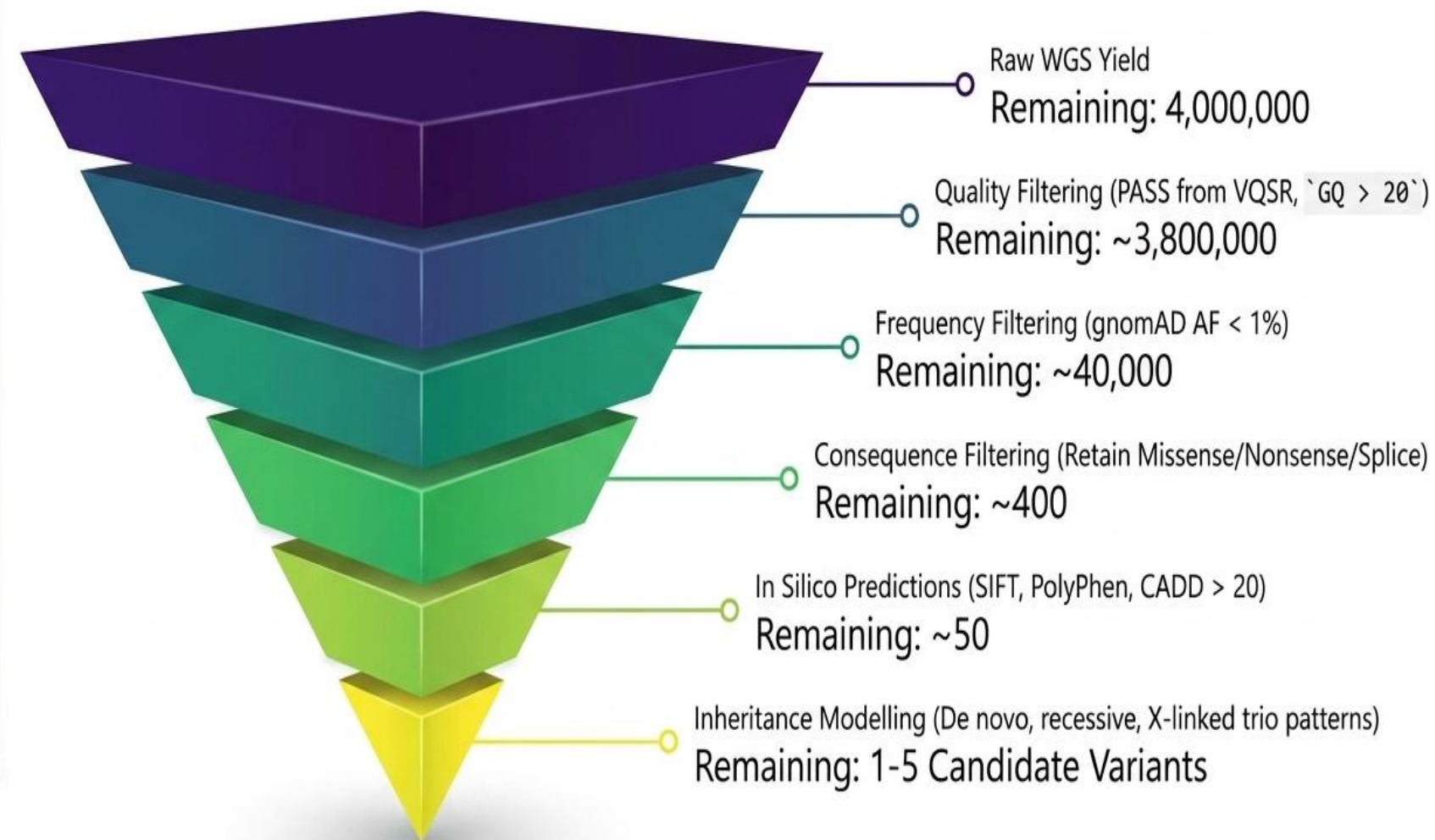
## [ INTERPRETATION ]

```
perl vcf2maf.pl \  
  --input-vcf $VEP_VCF \  
  --output-maf $OUT_MAF \  
  --tumor-id $SAMPLE \  
  --retain-info CADD_PHRED,gnomADg_AF
```

**Purpose:** The complex, highly nested VCF structure is notoriously difficult to ingest into R/Python analysis packages. We break out the VEP annotations into a flat, tabular Mutation Annotation Format (MAF) file for clinical reporting.

Input: VEP VCF

Output: Cohort MAF





# Reproducibility and Translational Impact

## Workflow Automation



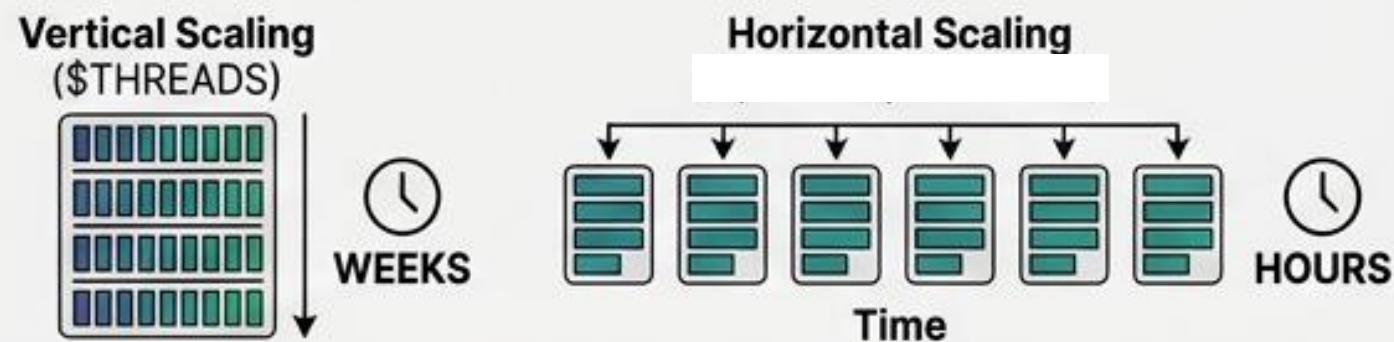
While bash scripts teach the logic, production pipelines mandate **Nextflow**, **Snakemake**, or Cromwell/WDL for parallelization and fault tolerance.

## Containerization



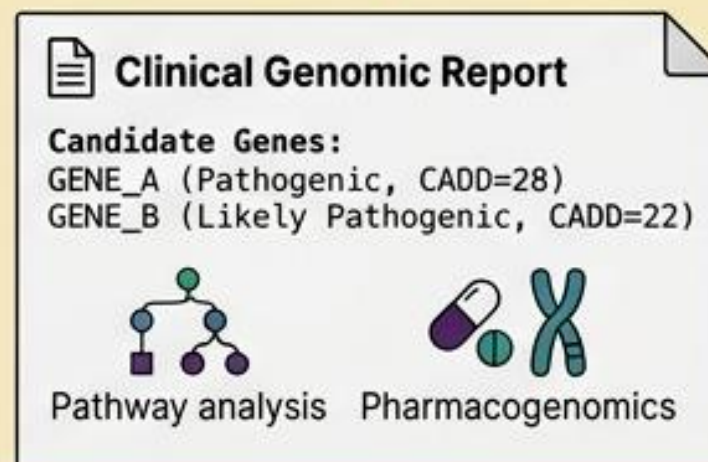
GATK and VEP computational environments must be version-locked via Docker or Singularity to ensure a genome analyzed in **2024** perfectly matches a re-analysis in 2030.

## Compute Economics



WGS requires massive compute. Spark and cluster parallelization reduce processing of 50GB+ files from weeks down to hours.

## The Clinical Endpoint



MAFs feed into pathway analysis and pharmacogenomics, ultimately resolving rare-disease diagnostic odysseys.